IIMT3642: Managing and Mining Big Data - Naive Bayes and Logistic Regression

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Outline

Naive Bayes Classification

Logistic Regression

K-fold Cross Validation

Reading for this lecture:

Data Science for Business, Foster Provost and Tom Fawcett, O'Reilly, 2013.
pp. 233 - 244, pp. 81-88, pp. 103-107,

CID	Balance	Age	Employed	Default
1	<50K	>=45	No	Yes
2	<50K	<45	Yes	No
3	>=50K	>=45	No	Yes
4	>=50K	>=45	Yes	No
5	>=50K	>=45	No	Yes
6	<50K	>=45	No	No
7	>=50K	<45	Yes	No
8	<50K	>=45	No	No
9	<50K	>=45	No	No
10	>=50K	<45	No	No
11	>=50K	<45	No	Yes
12	>=50K	<45	No	Yes
13	>=50K	<45	No	No
14	>=50K	<45	No	No
15	>=50K	<45	No	No

Bayes' Rule

- Let us consider some probabilities:
  - Prior Probabilities:
    - $P(Age >= 45) = \frac{7}{15} = 0.47$
    - $P(Balance < 50K) = \frac{5}{15} = 0.33$
  - The Joint Probability:
    - $P(Age >= 45, Balance < 50K) = \frac{4}{15} = 0.27$ when $Age >= 45$ and $Balance < 50K$ both satisfied
  - The Conditional Probability:
    - $P(Balance < 50K \mid Age >= 45) = \frac{4}{7} = 0.57$

We can observe that

- Given
  - prior probability $P(Age >= 45) = \frac{7}{15}$
  - Joint probability $P(Age >= 45, Balance < 50K) = \frac{4}{15}$
  - Conditional probability $P(Balance < 50K \mid Age >= 45) = \frac{4}{7}$
- The following relationship holds between the joint prob. and the conditional prob.
  - $P(Age >= 45, Balance < 50K) = P(Age >= 45) \times P(Balance < 50K \mid Age >= 45)$

Formally speaking,

- Considering two events A and B, we have
  - $P(AB) = P(A)P(B|A)$
  - $P(BA) = P(B)P(A|B)$
  - $P(A)P(B|A) = P(B)P(A|B)$
- The Bayes' Rule states that

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Feature Independence

- Given prior probabilities and the joint probability
  - $P(Age >= 45) = \frac{7}{15} = 0.47$
  - $P(Balance < 50K) = \frac{5}{15} = 0.33$
  - $P(Age >= 45, Balance < 50K) = \frac{4}{15} = 0.27$
- We observe that
  - $P(Age >= 45, Balance < 50K) \neq P(Age >= 45) \times P(Balance < 50K)$
- As a result, we know that the conditions ($Age >= 45$) and ($Balance < 50K$) are dependent

In contrast, if we roll a fair dice twice, it will provide

- Priors
  - $P(first\ roll\ =\ 1) = \frac{1}{6}$
  - $P(second\ roll\ =\ 2) = \frac{1}{6}$
- Joint probability
  - $P(first\ roll\ =\ 1\ , second\ roll\ =\ 2) = \frac{1}{36}$
- In this case, we observe
  - $P(first\ roll\ =\ 1\ , second\ roll\ =\ 2)\ =\ P(first\ roll\ =\ 1) \times P(second\ roll\ =\ 2)$
- So, we consider the conditions $(first\ roll\ =\ 1)$ and $(second\ roll\ =\ 2)$ are independent

Consider two event A and B, we have

- $P(A, B) = P(A)P(B|A)$

If event A and B are independent, we have

- $P(A, B) = P(A)\underline{P(B|A)} = P(A)\underline{P(B)}$

Probability Chain Rule

Given the Bayes' Rule, a chain rule of expanding joint probability of 3 events will hold:

$$P(A, B, C) = P(A|B, C)P(B|C)P(C)$$

This chain rule generalizes to joint probability with any number of events:

$$P(d_1, d_2, \dots, d_n)$$
$$= P(d_1|d_2, \dots, d_n)P(d_2|d_3, \dots, d_n) \dots P(d_{n-2}|d_{n-1}, d_n)P(d_{n-1}|d_n)P(d_n)$$

Naive Bayes Classification

The Naive Bayes classification problem is to find out the probability of class event when data are provided in many attributes.

For example,

Name	Balance	Age	Employed	Default
Jay	$120,000	42	No	?

$$P(\text{Default}|\text{Balance } \geq 50K, \text{Age} < 45, \text{ Employed} = \text{No})$$
$$= \frac{P(\text{Balance} \geq 50K, \text{Age} < 45, \text{Employed} = \text{No}|\text{Default})P(\text{Default})}{P(\text{Balance} \geq 50K, \text{Age} < 45, \text{Employed} = \text{No})}$$

Naive Bayes Classification is formulated and derived as follows.

- C: class attribute, e.g., <u>default</u> $(C = 1)$, <u>not default</u> $(C = 0)$
- $d_1, d_2, \ldots, d_n$: descriptive attributes

$$
\begin{aligned}
&P(C = 1 | d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n) \\
&= \frac{P(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n | C = 1) P(C = 1)}{P(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n)} \\
&= \frac{P(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n | C = 1) P(C = 1)}{P(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n, C = 1) + P(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n, C = 0)} \\
&= \frac{P(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n | C = 1) P(C = 1)}{P(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n | C = 1) P(C = 1) + P(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n | C = 0) P(C = 0)}
\end{aligned}
$$

Solving NB Classification

- Estimating $P(C = 1)$ and $P(C = 0)$ is convenient:
  - $P(C = 1)$ = fraction of instances with $C = 1$
  - $P(C = 0)$ = fraction of instances with $C = 0$
  - In the loan training example, we have
    - $P(C = 1) = 0.33$,
    - $P(C = 0) = 0.67$.

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Laplace Correction

- Smoothing is used to tackle the issue of zero probability
  - Helpful when any of the classes end up with 0 examples in the training data
- Laplace correction

$$P(C) = \frac{(\#\text{ of training instances belonging to class C}) + 1}{(\#\text{ of training instances}) + K}$$

In which, K is the number of predefined classes

- In our example, estimating $P(C = 1)$ and $P(C = 0)$ with Laplace correction yields
  - $P(C = 1) = 0.35$ and $P(C = 0) = 0.65$

- Estimating the conditional probabilities in the red boxes is extremely difficult:
  - Assuming every descriptive attribute is binary, the number of possible value combination for n terms $(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n)$ is 2^n
  - Some combinations may not even appear in the training data and hence there is no way to specify their probabilities of occurrence

$$
\begin{aligned}
&P(C=1|d_1=v_1,d_2=v_2,\ldots,d_n=v_n)\\
&=\frac{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n|C=1)P(C=1)}{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n)}\\
&=\frac{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n|C=1)P(C=1)}{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n,C=1)+P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n,C=0)}\\
&=\frac{\boxed{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n|C=1)}P(C=1)}{\boxed{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n|C=1)}P(C=1)+\boxed{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n|C=0)}P(C=0)}
\end{aligned}
$$

Naive Bayes Solution

- Conditional Independence Assumption
  - Assume descriptive attributes are independent of each other when class value C is given
  - This assumption might sacrifice prediction accuracy to some extent but solve the difficulty

$$P(d_1 = v_1, d_2 = v_2, \ldots, d_n = v_n | C = 1) P(C = 1)$$

$$= P(d_1 = v_1 | d_2 = v_2, \ldots, d_n = v_n, C = 1) P(d_2 = v_2 | d_3 = v_3, \ldots, d_n = v_n, C = 1) \ldots P(d_n = v_n | C = 1) P(C = 1)$$

$$= P(d_1 = v_1 | C = 1) P(d_2 = v_2 | C = 1) \ldots P(d_n = v_n | C = 1) P(C = 1)$$

Solution overview

$$P(C=1|d_1=v_1,d_2=v_2,\ldots,d_n=v_n)$$

$$=\frac{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n|C=1)P(C=1)}{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n)}$$

$$=\frac{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n|C=1)P(C=1)}{P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n|C=1)P(C=1)+P(d_1=v_1,d_2=v_2,\ldots,d_n=v_n|C=0)P(C=0)}$$

Apply Naive Bayes Assumption

$$=\frac{P(d_1=v_1|C=1)P(d_2=v_2|C=1)\ldots P(d_n=v_n|C=1)P(C=1)}{\sum_{k\in\{0,1\}}P(d_1=v_1|C=k)P(d_2=v_2|C=k)\ldots P(d_n=v_n|C=k)P(C=k)}$$

- Estimating $P(d_1 = v_1|C = 1), P(d_2 = v_2|C = 1), \ldots, P(d_n = v_n|C = 1)$ are fairly easy:
  - Assuming every descriptive attribute is binary, there are only n estimations.
  - We can estimate all probabilities from the training data

$$= \frac{\boxed{P(d_1 = v_1|C = 1)}P(d_2 = v_2|C = 1)\ldots P(d_n = v_n|C = 1)P(C = 1)}{\sum_{k\in\{0,1\}} P(d_1 = v_1|C = k)P(d_2 = v_2|C = k)\ldots P(d_n = v_n|C = k)P(C = k)}$$

For example, we can simply count to estimate $P(d_1 = v_1|C = 1)$ as the following fraction:

$$P(d_1 = v_1|C = 1) = \frac{\#\text{ of training instances with C=1 and } d_1 = v_1}{\#\text{ of training instance with C=1}}$$

- With Laplace correction, $P(d_1 = v_1 | C = 1)$ is estimated as

$$P(d_1 = v_1 | C = 1) = \frac{(\#\text{ of training instances with C=1 and } d_1 = v_1) + 1}{(\#\text{ of training instance with C=1}) + J}$$

In which, J is the number of possible values for the given descriptive attribute (e.g., d_1, NOT class).

Naive Bayes Classification

- Using the training data, we can estimate all components for Naive Bayes classification in the example.
  - Class attributes: $P(C = 1) = 0.35$; $P(C = 0) = 0.65$
  - Conditional Probabilities of descriptive attributes (i.e., Employed, Age, Balance):
    - Employed
      - When $C = 1$: $P(Employed = No|C = 1) = 0.86$, $P(Employed = Yes|C = 1) = 0.14$
      - When $C = 0$: $P(Employed = No|C = 0) = 0.67$, $P(Employed = Yes|C = 0) = 0.33$
    - Age
      - When $C = 1$: $P(Age >= 45 |C = 1) = 0.57$, $P(Age < 45|C = 1) = 0.43$
      - When $C = 0$: $P(Age >= 45 |C = 0) = 0.42$, $P(Age < 45|C = 0) = 0.58$
    - Balance
      - When $C = 1$: $P(Balance >= 50K |C = 1) = 0.71$, $P(Balance < 50K|C = 1) = 0.29$
      - When $C = 0$: $P(Balance >= 50K |C = 0) = 0.58$, $P(Balance < 50K|C = 0) = 0.42$

Name	Balance	Age	Employed	Default
Jay	$120,000	42	No	?

$$P(\text{Default} = 1 | \text{Balance } \geq 50K, \text{Age } < 45, \text{ Employed=No})$$

$$= \frac{P(\text{Balance } \geq 50K, \text{Age } < 45, \text{ Employed=No} | \text{Default}) P(\text{Default})}{P(\text{Balance } \geq 50K, \text{Age } < 45, \text{ Employed=No})}$$

- Compute for the numerator

$$P(\text{Balance } \geq 50K, \text{Age } < 45, \text{ Employed=No}|\text{Default})P(\text{Default})$$

$$= P(\text{Balance } \geq 50K, \text{Age } < 45, \text{ Employed=No}|C = 1)P(C = 1)$$

$$= P(\text{Balance } \geq 50K|C = 1)P(\text{Age } < 45|C = 1)P(\text{Employed=No}|C = 1)\, P(C = 1)$$

$$= 0.71 \times 0.43 \times 0.86 \times 0.35$$

$$= 0.09$$

- Compute for the denominator

$$P(\text{Balance } \geq 50K,\ \text{Age} < 45,\ \text{Employed=No})$$

$$= P(\text{Balance } \geq 50K,\ \text{Age} < 45,\ \text{Employed=No},\ C = 1) + P(\text{Balance } \geq 50K,\ \text{Age} < 45,\ \text{Employed=No}, C = 0)$$

$$= P(\text{Balance } \geq 50K,\ \text{Age} < 45,\ \text{Employed=No} | C = 1)P(C = 1) + P(\text{Balance } \geq 50K,\ \text{Age} < 45,\ \text{Employed=No} | C = 0)P(C = 0)$$

$$= 0.09 + P(\text{Balance } \geq 50K | C = 0)P(\text{Age} < 45 | C = 0)P(\text{Employed=No} | C = 0)P(C = 0)$$

$$= 0.09 + 0.58 \times 0.58 \times 0.67 \times 0.65$$

$$= 0.24$$

Naive Bayes Inference

Name	Balance	Age	Employed	Default
Jay	$120,000	42	No	?

$$P(\text{Default} = 1 | \text{Balance } \geq 50K, \text{Age } < 45, \text{ Employed=No})$$

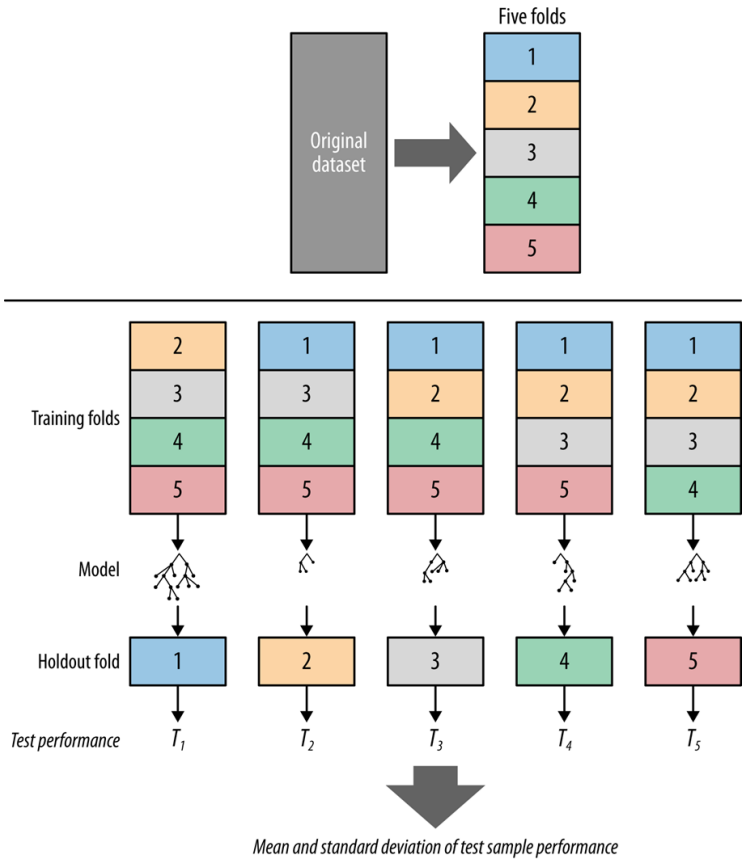
$$= \frac{P(\text{Balance } \geq 50K, \text{Age } < 45, \text{ Employed=No} | \text{Default}) P(\text{Default})}{P(\text{Balance } \geq 50K, \text{Age } < 45, \text{ Employed=No})}$$

$$= \frac{0.09}{0.24} = 0.375 < 0.5$$

No Default

Method Selection

- K-fold Cross Validation
  - Choose between different models, based on their performance
- Select the method with the highest mean validation performance
- Apply the selected method to the whole data (e.g., all five partitions)



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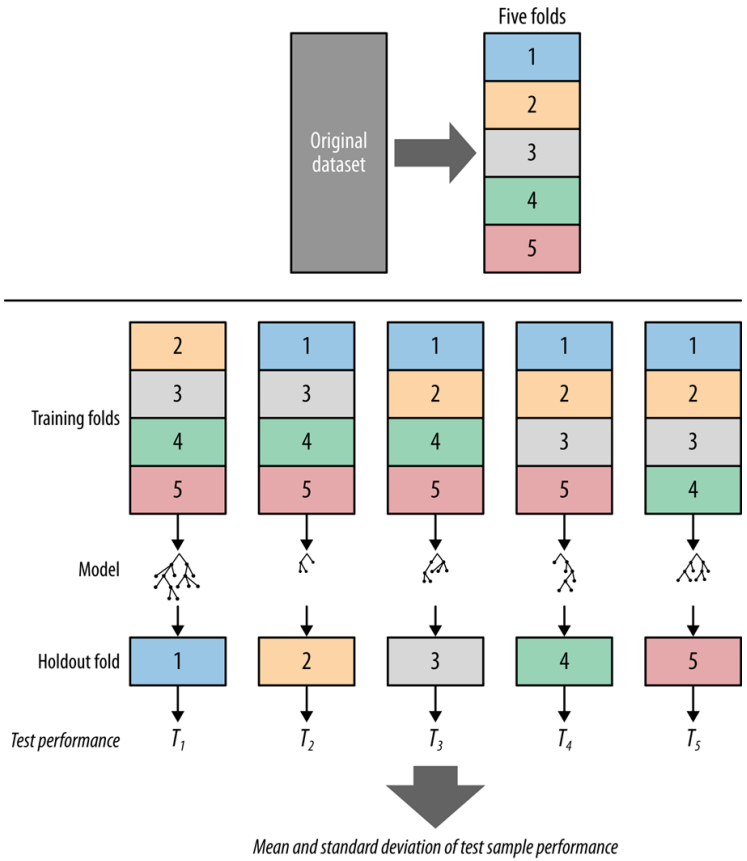
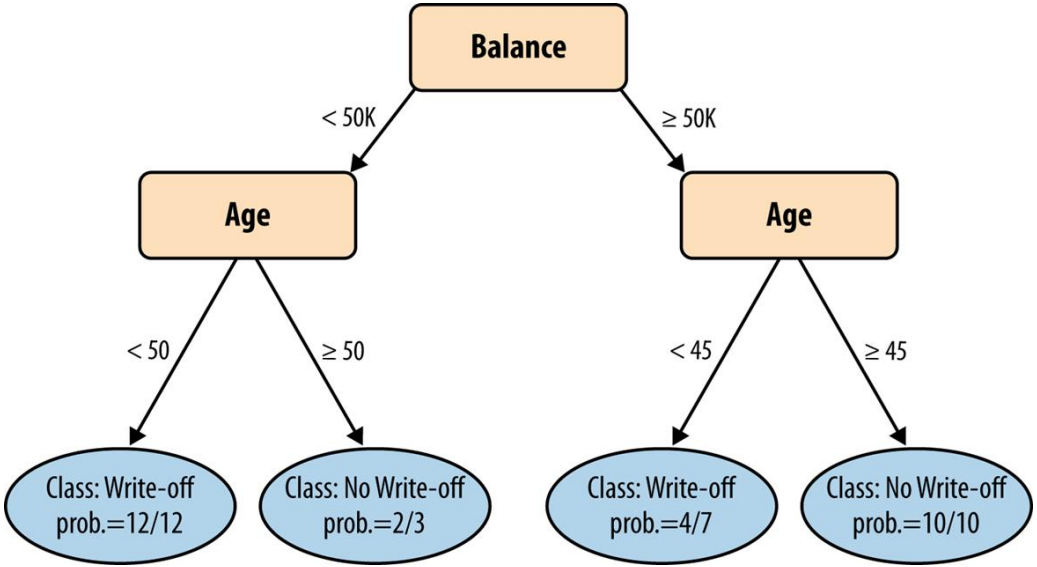


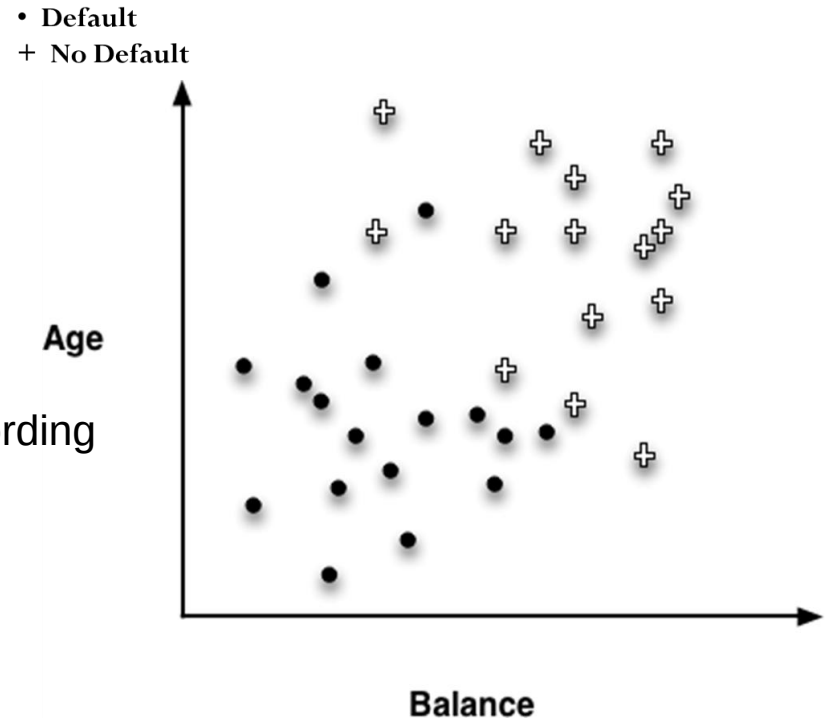
Linear Models

- Decision Tree Model
  - Branching over attributes
  - Could be viewed as a partitioning of data



Linear Models

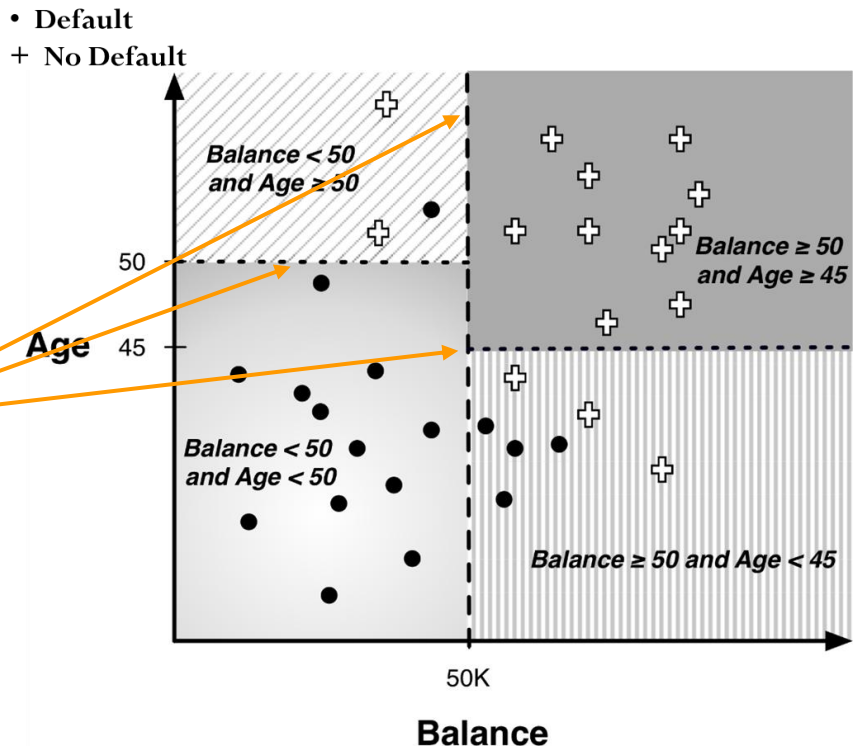
- Decision Tree Model
- Data View



Instance space: plotting training instances according to their values of descriptive attributes.

- Decision Tree Model
- Data View
- Decision Boundary

A decision tree model is essentially a set of horizontal/vertical decision boundaries that partition a instance space into groups of homogeneous instances.

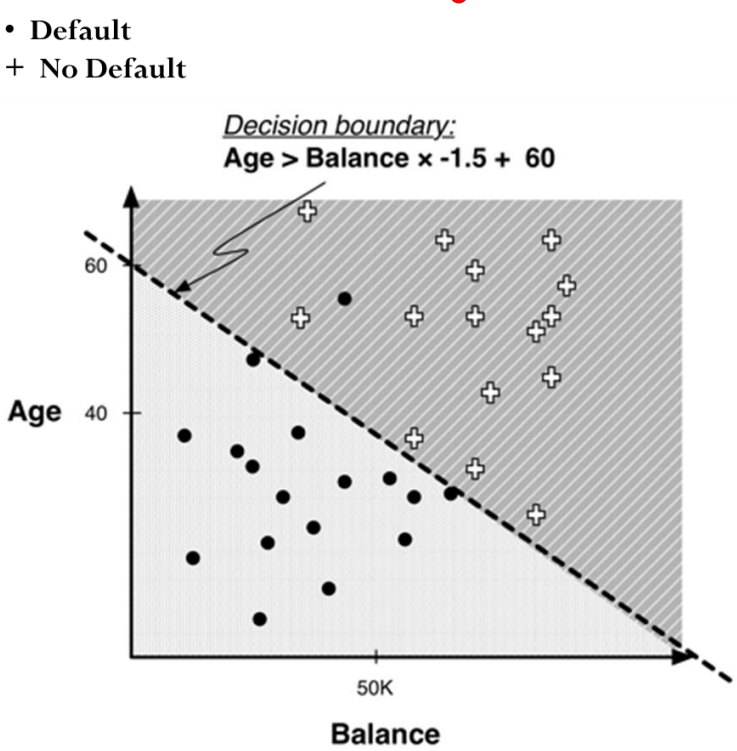


- Decision Tree Model
- Data View
- Decision Boundary
- Linear Model
  - Linear decision boundary

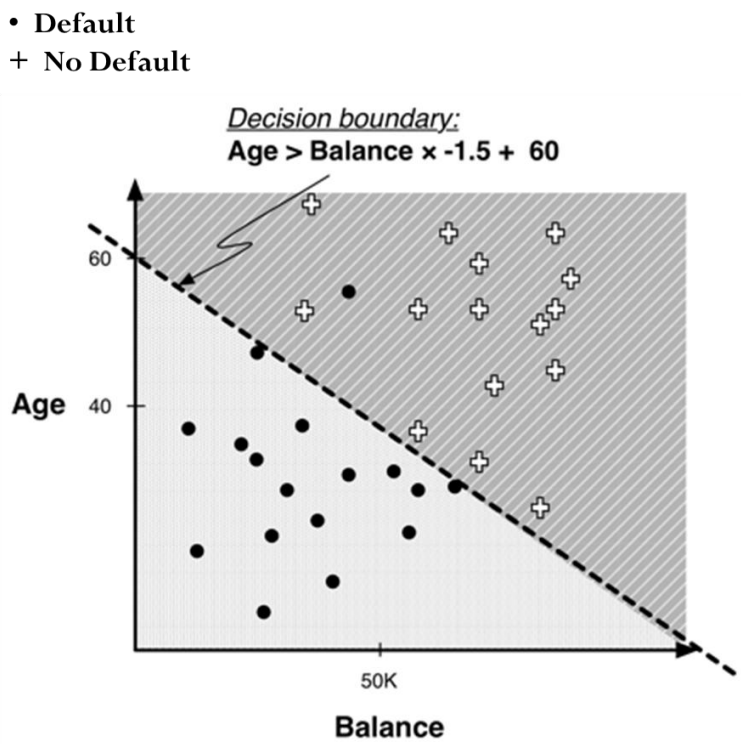
Decision Boundary:

$$w_0 + w_1 \times Age + w_2 \times Balance = 0$$

$$-60 + Age + 1.5 \times Balance = 0$$



- Decision Tree Model
- Data View
- Decision Boundary
- Linear Model
- Classification



Classify instances using the decision boundary:

$$+ \quad \text{if } -60 + Age + 1.5 \times Balance > 0$$

$$\circ \quad \text{if } -60 + Age + 1.5 \times Balance \leq 0$$

- A linear model employs a linear combination of descriptive attributes, namely decision boundary or $f(x)$, to classify instances.

$$f(\boldsymbol{x}) = w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n$$

In which,

  - The values of coefficient weight $w_0, w_1, w_2, \ldots, w_n$ are learned from training data.
  - A new instance (not in the training data) is classified as one class if $f(\boldsymbol{x}) > 0$ or as the other class if $f(\boldsymbol{x}) \leq 0$
  - Absolute values of $w_1, w_2, \ldots w_n$ indicate the importance of their respectively associated descriptive attribute in classifying an instance.
  - The larger the absolute value of w_i, the more important its associated attribute in classification.

Logistic Regression

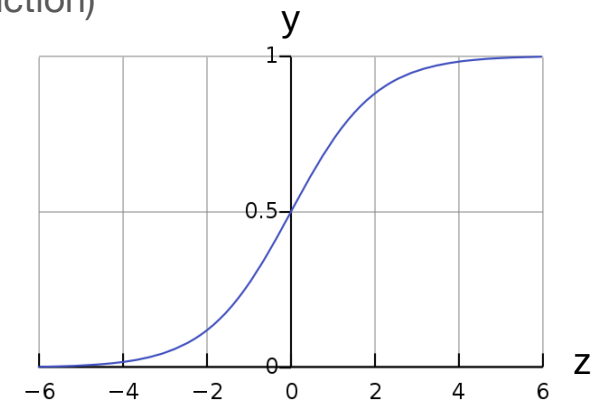
- Logistic regression learns from training data to predict the probability $P(Y = C|\boldsymbol{x})$.
  - Y: response variable
  - C: class attribute values, e.g., default or no default
  - $\boldsymbol{x} = (x_1, x_2, \ldots, x_n)$: a set of descriptive attributes, e.g., age, balance, employed

For example, we are interested in finding out:

$$P(Y = Default|Age < 45, Balance >= 50K, Employed = No)$$

- Logistic Regression assumes a logistic function (a.k.a. sigmoid function)

$$P(Y=1|\boldsymbol{x}) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-(w_1x_1+w_2x_2+\cdots+w_nx_n)}}$$



and

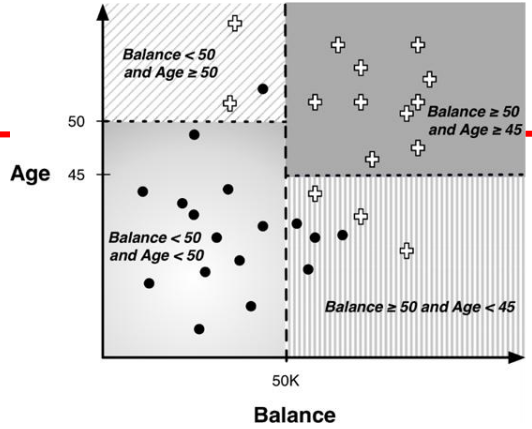
$$P(Y=0|\boldsymbol{x}) = 1 - P(Y=1|\boldsymbol{x}) = \frac{e^{-(w_1x_1+w_2x_2+\cdots+w_nx_n)}}{1+e^{-(w_1x_1+w_2x_2+\cdots+w_nx_n)}} = \frac{1}{1+e^{(w_1x_1+w_2x_2+\cdots+w_nx_n)}}$$

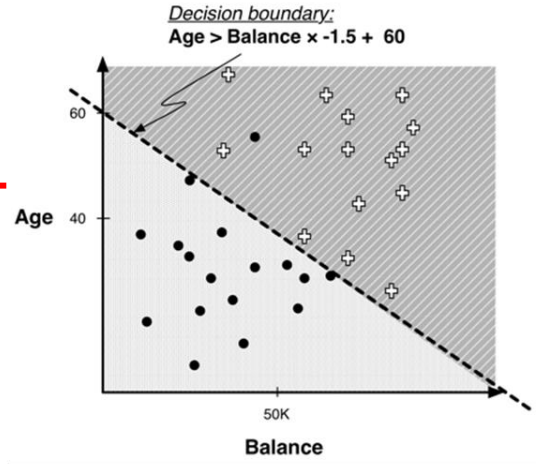
- Logistic regression will
  - Learn values of $w_0, w_1, w_2, \ldots, w_n$ from training data
  - Then, predict $P(Y = 1|\boldsymbol{x})$ and $P(Y = 0|\boldsymbol{x})$ for a new instance using the equations in the previous slide
  - Classify a new instance as class 1 if $P(Y = 1|\boldsymbol{x}) > P(Y = 0|\boldsymbol{x})$ or class 0 otherwise

Method Comparison

- Functionality
  - All three methods (decision tree, Naïve Bayes, logistic regression) can classify a new instance and predict the probability that a new instance belonging to a class.
- Performance
  - Use cross-validation to evaluate the performance of each method.
- Methodology
- Comprehensibility
- Explanatory Power

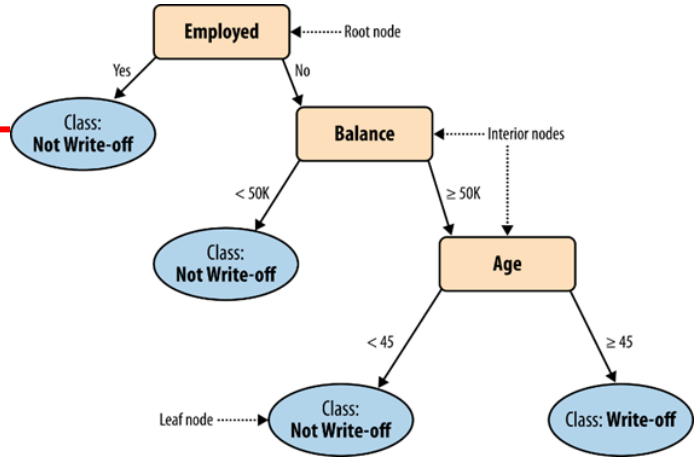
Method Comparison





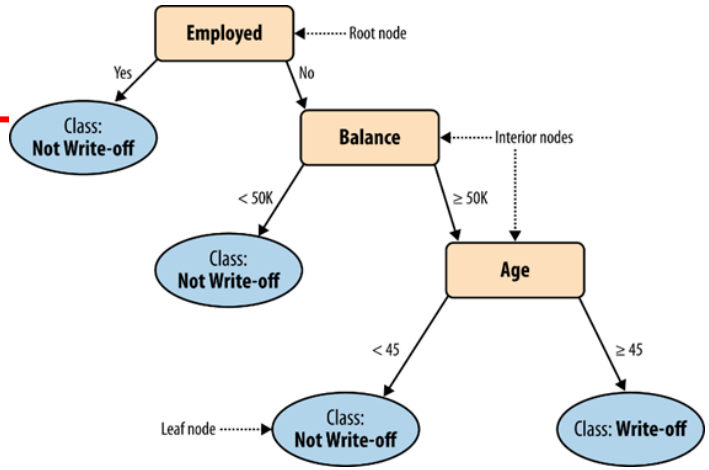
- Functionality
- Performance
- Methodology
  - Decision tree is a piecewise classifier (i.e., pick one attribute at a time) and create many decision boundaries to partition the instance space.
  - Logistic regression considers all attributes at the same time (i.e., a linear combination of attributes) and create one decision boundary to partition a instance space
  - Naïve Bayes is not a partition approach but employs Bayes rule and conditional independence assumption to estimate the probability a new instance belonging to a class.
- Comprehensibility
- Explanatory Power

Method Comparison



- Functionality
- Performance
- Methodology
- Comprehensibility
  - Decision tree visualize the produced model and it is easy to be understood by managers without strong background in statistics and data mining. However, tree may grow too large in size.
  - Logistic regression is easy to understand for managers with background in statistics.
  - For managers, Naïve Bayes only produces a probability estimation and functions like a black box.
- Explanatory Power

Method Comparison



- Functionality
- Performance
- Methodology
- Comprehensibility
- Explanatory Power
  - Decision tree has good explanatory power. Generally, attributes appearing in higher levels of a tree are more important for classification than those in lower levels.
  - Logistic regression has strong explanatory power. Absolute values of w_1 , w_2 , … w_n indicate the importance of their respectively associated descriptive attribute in classifying an instance.
  - For managers, the explanatory power of Naïve Bayes is low.